

BLM5121 – Web Mining — Term Project Proposal

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WHY – Motivation

Academic collaboration networks are a cornerstone of Web Structure Mining, offering a dynamic view into how research communities form and evolve. Predicting future co-authorship links is not just an academic exercise; it has practical value in building research-partner recommendation systems and identifying emerging trends. For this project, I have chosen the **ogbl-collab** dataset from the Open Graph Benchmark. It provides a large-scale, real-world environment to benchmark three approaches of increasing complexity – from a classical structural heuristic to a deep graph neural network – which directly addresses the core challenges of this course.

WHAT – Problem & Algorithms

I have chosen the **ogbl-collab** dataset, which contains 235,868 authors and approximately 2.36 million directed collaboration edges (1.18 million undirected). The dataset features a time-based split: training edges up to 2017, validation edges from 2018, and test edges from 2019, simulating a realistic forecasting scenario. Each node is represented by a 128-dimensional feature vector obtained by averaging the word2vec embeddings of papers published by that author. The core task is **Link Prediction**: given a snapshot of the collaboration network up to a specific year, predict which new author-author edges will appear in the following year. *Source*: <https://ogb.stanford.edu/docs/linkprop/#ogbl-collab>

To approach this problem, I will compare three methods of increasing complexity:

- **Method 1: Common Neighbors** A classical structural heuristic that scores each candidate author pair by the number of collaborators they share. It requires no learning and no node features, operating solely on graph topology. This serves as a lower-bound reference to measure the added value of the subsequent learned methods.
- **Method 2: MLP (Multi-Layer Perceptron)** A purely feature-based learned method. For each candidate edge the pre-computed 128-dimensional node features are combined via element-wise multiplication and passed through a fully-connected network to produce a link probability score. Importantly, this method ignores graph structure entirely and learns only from node content.
- **Method 3: GCN (Graph Convolutional Network)** A graph neural network that refines node representations by iteratively aggregating information from each node's neighborhood before computing link scores. Unlike the MLP, the GCN exploits both graph topology and node features simultaneously, representing the modern deep learning approach to this problem.

HOW – Evaluation & Implementation

To ensure a fair comparison, all three methods will be evaluated conditions and the primary metric is **Hits@50**: for each true test collaboration, the model must rank it above at least 100,000 randomly sampled negative pairs for it to count as a hit. I also plan to analyze the trade-offs between performance and computational cost by tracking runtime and memory usage at 10%, 50%, and 100% of the dataset size. The implementation will be done in Python. To make the results accessible and interactive, I will build a GUI using Streamlit with three main screens: Dataset Manager, Algorithm Runner, Results Dashboard. The experiments will be executed on my own computer with an RTX 5060 Ti GPU.

WHEN – Milestone Timetable

I have planned the project execution according to the following schedule:

- **Weeks 1–2:** Dataset loading, preprocessing, and initial EDA; implement Method 1 (Common Neighbors) and verify the Hits@50 score using the official OGB evaluator.
- **Weeks 3–4:** Implement Method 2 (MLP) and Method 3 (GCN); integrate node features and adjacency information where applicable; tune key hyperparameters (learning rate, hidden dimensions, number of layers, dropout).
- **Week 5:** Build the Streamlit results dashboard, generate comparison charts across all three methods and dataset sizes, and write the final report for submission.